Manage NIS configuration

ONTAP 9.14.1 REST API reference

NetApp
February 12, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-restapi/ontap/security_authentication_cluster_nis_endpoint_overview.html on February 12, 2024. Always check docs.netapp.com for the latest.

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Manage NIS configuration

Security authentication cluster NIS endpoint overview

Overview

NIS servers are used to authenticate user and client computers. NIS domain name and NIS server information is required to configure NIS. This API retrieves and manages NIS server configurations.

Examples

Retrieving cluster NIS information

The cluster NIS GET request retrieves the NIS configuration of the cluster.

The following example shows how a GET request is used to retrieve the cluster NIS configuration:

```
# The API:
/security/authentication/cluster/nis

# The call:
curl -X GET "https://<mgmt-ip>/api/security/authentication/cluster/nis" -H
"accept: application/hal+json"

# The response:
{
"domain": "domainA.example.com",
"servers": [
  "10.10.10.10",
  "example.com"
],
"bound_servers": [
  "10.10.10.10"
]
}
```

Creating the cluster NIS configuration

The cluster NIS POST request creates a NIS configuration for the cluster.

The following example shows how a POST request is used to create a cluster NIS configuration:

```
# The API:
/security/authentication/cluster/nis

# The call:
curl -X POST "https://<mgmt-ip>/api/security/authentication/cluster/nis"
-H "accept: application/json" -H "Content-Type: application/json" -d "{
\"domain\": \"domainA.example.com\", \"servers\": [
\"10.10.10.10\",\"example.com\" ]}"
```

Updating the cluster NIS configuration

The cluster NIS PATCH request updates the NIS configuration of the cluster.

The following example shows how to update the domain:

```
# The API:
/security/authentication/cluster/nis

# The call:
curl -X PATCH "https://<mgmt-ip>/api/security/authentication/cluster/nis"
-H "accept: application/json" -H "Content-Type: application/json" -d "{
\"domain\": \"domainC.example.com\", \"servers\": [ \"13.13.13.13\" ]}"
```

The following example shows how to update the server:

```
# The API:
/security/authentication/cluster/nis

# The call:
curl -X PATCH "https://<mgmt-ip>/api/security/authentication/cluster/nis"
-H "accept: application/json" -H "Content-Type: application/json" -d "{
\"servers\": [ \"14.14.14.14\" ]}"
```

Deleting the cluster NIS configuration

The cluster NIS DELETE request deletes the NIS configuration of the cluster.

The following example shows how a DELETE request is used to delete the cluster NIS configuration:

```
# The API:
/security/authentication/cluster/nis

# The call:
curl -X DELETE "https://<mgmt-ip>/api/security/authentication/cluster/nis"
-H "accept: application/hal+json"
```

Delete the NIS configuration for the cluster

DELETE `/security/authentication/cluster/nis`

Introduced In: 9.6

Deletes the NIS configuration of the cluster. NIS can be removed as a source from ns-switch if NIS is not used for lookups.

Response

```
Status: 200, Ok
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve the NIS configuration for the cluster

GET `/security/authentication/cluster/nis`

Introduced In: 9.6

Retrieves the NIS configuration of the cluster. Both NIS domain and servers are displayed by default. The `bound_servers` property indicates the successfully bound NIS servers.

Parameters

Name	Type	In	Required	Description
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
binding_details	array[binding_details]	An array of objects where each object represents the NIS server and it's status for a given NIS domain. It is an advanced field.
bound_servers	array[string]	
domain	string	The NIS domain to which this configuration belongs.
servers	array[string]	A list of hostnames or IP addresses of NIS servers used by the NIS domain configuration.

Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "binding_details": {
  },
  "bound_servers": {
  },
  "servers": {
  }
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

binding_status

Name	Type	Description
code	string	Code corresponding to the server's binding status.
message	string	Detailed description of the server's binding status.

binding_details

Name	Type	Description
server	string	Hostname/IP address of the NIS server in the domain.
status	binding_status	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code

Name	Type	Description
message	string	Error message
target	string	The target parameter that caused the error.

Update the NIS configuration for the cluster

PATCH `/security/authentication/cluster/nis`

Introduced In: 9.6

Both NIS domain and servers can be updated. Domains and servers cannot be empty. Both FQDNs and IP addresses are supported for the 'servers' field. If the domain is updated, NIS servers must also be specified. IPv6 must be enabled if IPv6 family addresses are specified for the `servers` property.

Request Body

Name	Type	Description
_links	_links	
binding_details	array[binding_details]	An array of objects where each object represents the NIS server and it's status for a given NIS domain. It is an advanced field.
bound_servers	array[string]	
domain	string	The NIS domain to which this configuration belongs.
servers	array[string]	A list of hostnames or IP addresses of NIS servers used by the NIS domain configuration.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "binding_details": {
  },
  "bound_servers": {
  },
  "servers": {
  }
}
```

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
1966253	IPv6 is not enabled in the cluster .
3276964	The NIS domain name or NIS server domain is too long. The maximum supported for domain name is 64 characters and the maximum supported for NIS server domain is 255 characters.
3276933	A maximum of 10 NIS servers can be configured per SVM.
23724109	DNS resolution failed for one or more specified servers.
23724112	DNS resolution failed due to an internal error. Contact technical support if this issue persists.
23724132	DNS resolution failed for all the specified servers.

Error Code	Description
23724130	Cannot use an IPv6 name server address because there are no IPv6 interfaces

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

binding_status

Name	Type	Description
code	string	Code corresponding to the server's binding status.
message	string	Detailed description of the server's binding status.

binding_details

Name	Type	Description
server	string	Hostname/IP address of the NIS server in the domain.
status	binding_status	

cluster_nis_service

Name	Type	Description
_links	_links	
binding_details	array[binding_details]	An array of objects where each object represents the NIS server and it's status for a given NIS domain. It is an advanced field.
bound_servers	array[string]	
domain	string	The NIS domain to which this configuration belongs.
servers	array[string]	A list of hostnames or IP addresses of NIS servers used by the NIS domain configuration.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create the NIS configuration for the cluster

POST `/security/authentication/cluster/nis`

Introduced In: 9.6

The cluster can have one NIS server configuration. Specify the NIS domain and NIS servers as input. The servers field cannot be empty. Both FQDNs and IP addresses are supported for the `server` property. IPv6 must be enabled if IPv6 family addresses are specified in the `server` property. A maximum of ten NIS servers are supported.

Required properties

- `domain` - NIS domain to which this configuration belongs.
- `servers` - List of hostnames or IP addresses of NIS servers used by the NIS domain configuration.

Request Body

Name	Type	Description
_links	_links	
binding_details	array[binding_details]	An array of objects where each object represents the NIS server and it's status for a given NIS domain. It is an advanced field.

Name	Type	Description
bound_servers	array[string]	
domain	string	The NIS domain to which this configuration belongs.
servers	array[string]	A list of hostnames or IP addresses of NIS servers used by the NIS domain configuration.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "binding_details": {
  },
  "bound_servers": {
  },
  "servers": {
  }
}
```

Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of NIS domain records.
records	array[cluster_nis_service]	

Example response

```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "binding_details": {
    },
    "bound_servers": {
    },
    "servers": {
    }
  }
}
```

Headers

Name	Description	Type
Location	Useful for tracking the resource location	string

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
1966253	IPv6 is not enabled in the cluster.

Error Code	Description
3276964	The NIS domain name or NIS server domain is too long. The maximum supported for domain name is 64 characters and the maximum supported for NIS server domain is 255 characters.
3276933	A maximum of 10 NIS servers can be configured per SVM.
13434916	The SVM is in the process of being created. Wait a few minutes, and then try the command again.
23724109	DNS resolution failed for one or more specified servers.
23724112	DNS resolution failed due to an internal error. Contact technical support if this issue persists.
23724132	DNS resolution failed for all the specified servers.
23724130	Cannot use an IPv6 name server address because there are no IPv6 interfaces.
23724111	Invalid value specified for nis-servers. The value must be a valid hostname or IP address.
23724087	The specified IPv6 address is not supported because it is one of the following: ::, link-local, multicast, v4-compatible, v4-mapped, loopback.
23724086	The specified IPv4 address is not supported because it is one of the following: multicast, loopback, 0.0.0.0 or broadcast.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

binding_status

Name	Type	Description
code	string	Code corresponding to the server's binding status.
message	string	Detailed description of the server's binding status.

binding_details

Name	Type	Description
server	string	Hostname/IP address of the NIS server in the domain.
status	binding_status	

cluster_nis_service

Name	Type	Description
_links	_links	
binding_details	array[binding_details]	An array of objects where each object represents the NIS server and it's status for a given NIS domain. It is an advanced field.
bound_servers	array[string]	
domain	string	The NIS domain to which this configuration belongs.
servers	array[string]	A list of hostnames or IP addresses of NIS servers used by the NIS domain configuration.

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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